The Pulickal Algorithm: A Multi-Directional Block-Parallel Elimination Framework for Dense Linear Systems

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Abstract

Traditional direct solvers for dense linear systems, such as Gaussian elimination and LU decomposition, are inherently bottlenecked by strict sequential dependencies. The critical path of standard elimination forces a unidirectional data flow from the top-left to the bottom-right of the coefficient matrix, severely limiting parallel scalability on modern multi-core and distributed architectures. This paper introduces the **Pulickal Algorithm**, a novel multi-directional elimination framework that shatters this sequential dependency. By utilizing a bi-directional twisted block factorization, the algorithm initializes concurrent execution fronts at opposite polarities of the matrix. These independent fronts perform localized triangularizations and boundary updates that converge simultaneously on a centralized reduced system, termed the Pulickal Core. We outline the mathematical formulation of this dual-Schur domain decomposition, detail the algorithmic architecture, and provide a theoretical analysis of its computational complexity. This framework offers a highly scalable alternative for high-performance computing, maximizing parallel throughput and minimizing idle processor time during the dense factorization phase.

1 Introduction

The resolution of large, dense systems of linear equations, $Ax = b$, remains a cornerstone operation in high-performance scientific computing, computational fluid dynamics, and machine learning optimizations. While the asymptotic time complexity of direct methods remains bounded at $O(n^3)$, modern optimizations have largely focused on maximizing cache-hit ratios via Level-3 BLAS operations (e.g., Block LU factorization).

However, these traditional block methods remain hardware-constrained by a fundamental Directed Acyclic Graph (DAG) of dependencies. In standard Block LU, the factorization of the k-th block strictly depends on the completion of the $(k-1)$-th block. This unidirectional propagation creates a "tail effect," where parallel efficiency drops significantly towards the end of the factorization as the remaining sub-matrix shrinks and processors are left idle.

The **Pulickal Algorithm** proposes a structural paradigm shift: abandoning unidirectional elimination in favor of a concurrent, multi-directional approach. By treating both the top-left and bottom-right extremities of the matrix as independent, initial computational fronts,

we halve the critical path of the elimination DAG. This bi-directional approach proves that multiple elimination fronts can safely converge on a central block, exposing a massive degree of coarse-grained parallelism previously inaccessible to standard direct solvers.

2 Mathematical Formulation: The Pulickal Decomposition

Consider a non-singular, dense, square matrix $A \in \mathbb{R}^{n \times n}$ and a system $Ax = b$. To establish the multi-directional fronts, we partition A, x, and b into a 3×3 block structure. Let the dimensions of the blocks be p, r, and q respectively, such that $p + r + q = n$.

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \tag{1}$$

In the Pulickal Algorithm, $A_{11} \in \mathbb{R}^{p \times p}$ represents the **Alpha Front** (top-left) and $A_{33} \in \mathbb{R}^{q \times q}$ represents the **Beta Front** (bottom-right). The central block $A_{22} \in \mathbb{R}^{r \times r}$ acts as the convergence zone.

2.1 Concurrent Front Elimination

Unlike standard solvers, we assume A_{11} and A_{33} are non-singular and factor them simultaneously:

- **Alpha Factorization:** $A_{11} = L_\alpha U_\alpha$
- **Beta Factorization:** $A_{33} = U_\beta L_\beta$ *(Note: Beta utilizes a reverse-order LU to eliminate upwards).*

Because A_{11} and A_{33} are disjoint, the transformation matrices required to eliminate the off-diagonal column blocks commute. They can be applied entirely in parallel without race conditions.

2.2 The Pulickal Core (Dual Schur Complement)

Applying the Alpha and Beta eliminations to the central row block yields the reduced system for x_2. We define this convergence state as the **Pulickal Core** (S_P), a generalization of the Schur complement subjected to bi-directional updates:

$$S_P = A_{22} - A_{21}A_{11}^{-1}A_{12} - A_{23}A_{33}^{-1}A_{32} \tag{2}$$

The corresponding right-hand side vector concurrently converges to:

$$\hat{b}_2 = b_2 - A_{21}A_{11}^{-1}b_1 - A_{23}A_{33}^{-1}b_3 \tag{3}$$

The solution for the central block is $S_P x_2 = \hat{b}_2$. Once x_2 is resolved, the algorithm executes a coupled bi-directional back-substitution to solve for the remaining variables.

3 Algorithmic Architecture

The Pulickal Algorithm is structured into four distinct, highly parallelizable phases, optimized for distributed memory systems.

Algorithm 1 The Pulickal Algorithm: Multi-Directional Block Elimination

Require: Matrix A, Vector b, Block dimensions p, r, q

Phase 1: Bi-Frontal Asynchronous Factorization

Group A: Factor $A_{11} = L_\alpha U_\alpha$ and forward substitute b_1

Group B: Factor $A_{33} = U_\beta L_\beta$ and forward substitute b_3

Phase 2: Matrix Cross-Update (The Convergence)

Group A: Compute left-side updates $W_L = A_{21}A_{11}^{-1}A_{12}$

Group B: Compute right-side updates $W_R = A_{23}A_{33}^{-1}A_{32}$

Assemble Pulickal Core globally: $S_P = A_{22} - W_L - W_R$

Assemble $\hat{b}_2 = b_2 - A_{21}A_{11}^{-1}b_1 - A_{23}A_{33}^{-1}b_3$

Phase 3: Core Resolution

Solve the reduced system $S_P x_2 = \hat{b}_2$ using robust partial pivoting

Broadcast x_2 back to all processor groups.

Phase 4: Consistent Multi-Directional Back-Substitution

Compute updated boundary vectors $\tilde{b}_1 = b_1 - A_{12}x_2$ and $\tilde{b}_3 = b_3 - A_{32}x_2$

Solve the coupled 2×2 block system for (x_1, x_3):

$$\begin{bmatrix} A_{11} & A_{13} \\ A_{31} & A_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} \tilde{b}_1 \\ \tilde{b}_3 \end{bmatrix}$$

return $x = [x_1, x_2, x_3]^T$

4 Numerical Stability and Pulickal Pivoting

A known limitation of standard domain-decomposition methods is the breakdown of global partial pivoting. Because the Alpha and Beta fronts are factored independently prior to core assembly, traditional global pivoting is impossible without forcing synchronization.

To maintain numerical stability, we propose **Localized Pulickal Pivoting**. Pivoting is strictly confined within the boundaries of A_{11} and A_{33}. To prevent the amplification of rounding errors when constructing the Pulickal Core S_P, the algorithm employs a dynamic block-sizing heuristic. If a localized pivot falls below a predefined tolerance threshold ϵ, the boundary of the block is dynamically shifted. This effectively expands the dimension r of the central core A_{22}, deferring ill-conditioned rows to Phase 3, where standard robust pivoting can be safely applied to S_P.

5 Complexity and Parallel Scalability

While the total arithmetic volume of the Pulickal Algorithm remains bounded by $O(n^3) \approx \frac{2}{3}n^3$ FLOPs (preserving the theoretical efficiency of standard Gaussian elimination), the primary architectural breakthrough lies in the **critical path length**.

Let $T(N)$ be the wall-clock time required to factor an $N \times N$ matrix. In standard block

solvers, the critical path scales linearly with the number of blocks. In the Pulickal framework, assuming balanced partitioning ($p \approx q \approx \frac{n}{2}$) and a relatively small core r, the theoretical factorization bottleneck is reduced to:

$$T_{Pulickal} \approx \max(T(p), T(q)) + T(r) + T_{comm} \tag{4}$$

Because p and q are processed concurrently, the execution time approaches $T(n/2) + T_{comm}$. By enabling simultaneous elimination from multiple fronts, this algorithm successfully halves the wall-clock time required for the factorization phase on distributed systems, directly translating to higher processor utilization and reduced time-to-solution.

6 Conclusion and Future Work

We have established the mathematical and algorithmic foundations of the Pulickal Algorithm. By abandoning the strictly sequential, unidirectional constraints of traditional LU decomposition in favor of a bi-directional twisted factorization, this framework exposes significant coarse-grained parallelism and enables up to a $2\times$ reduction in critical path length.

Future work will focus on empirical validation. We intend to implement the Pulickal Algorithm utilizing MPI and OpenMP to benchmark its real-world wall-clock speedup, measure the overhead of dynamic pivoting boundary adjustments, and compare its operational throughput against standard ScaLAPACK routines on enterprise supercomputing clusters.